

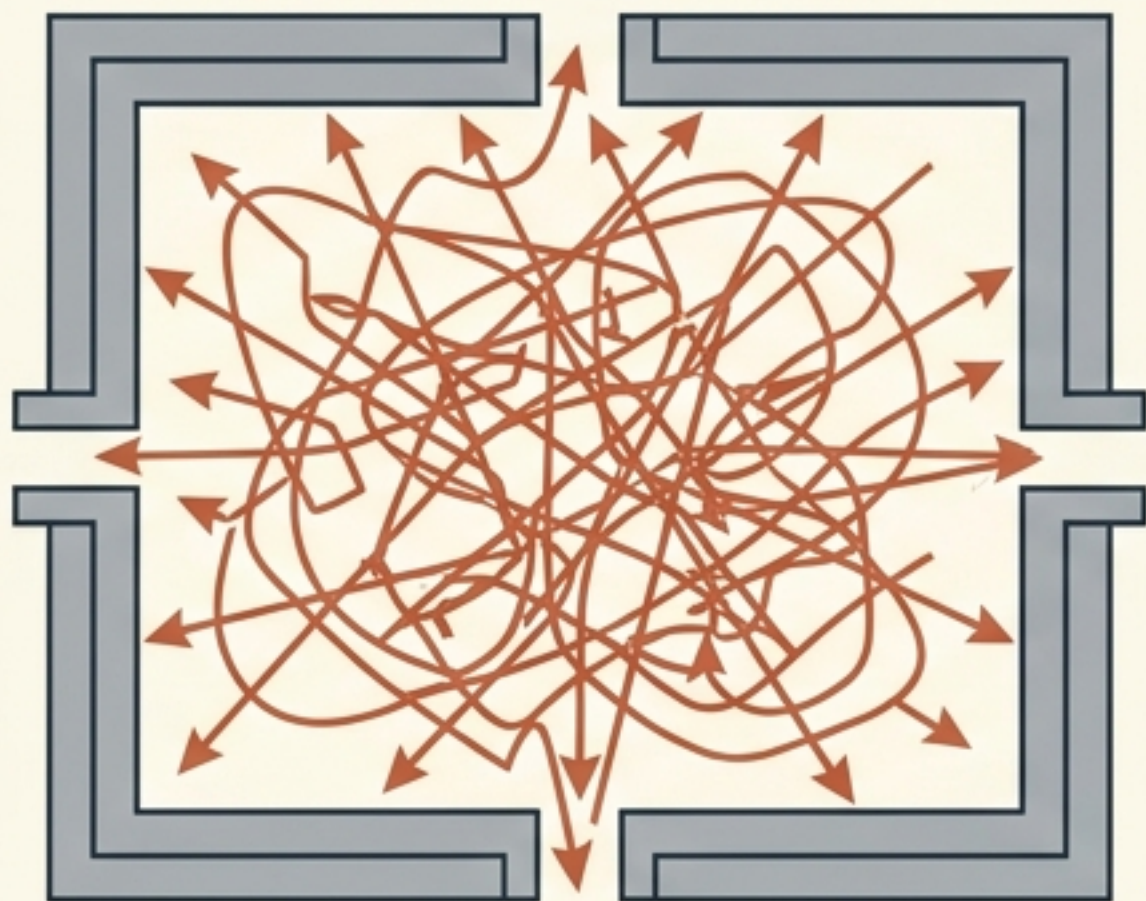
The Paradox of Precaution

How engineering perfect AGI safety mathematically guarantees the collapse of human trust.

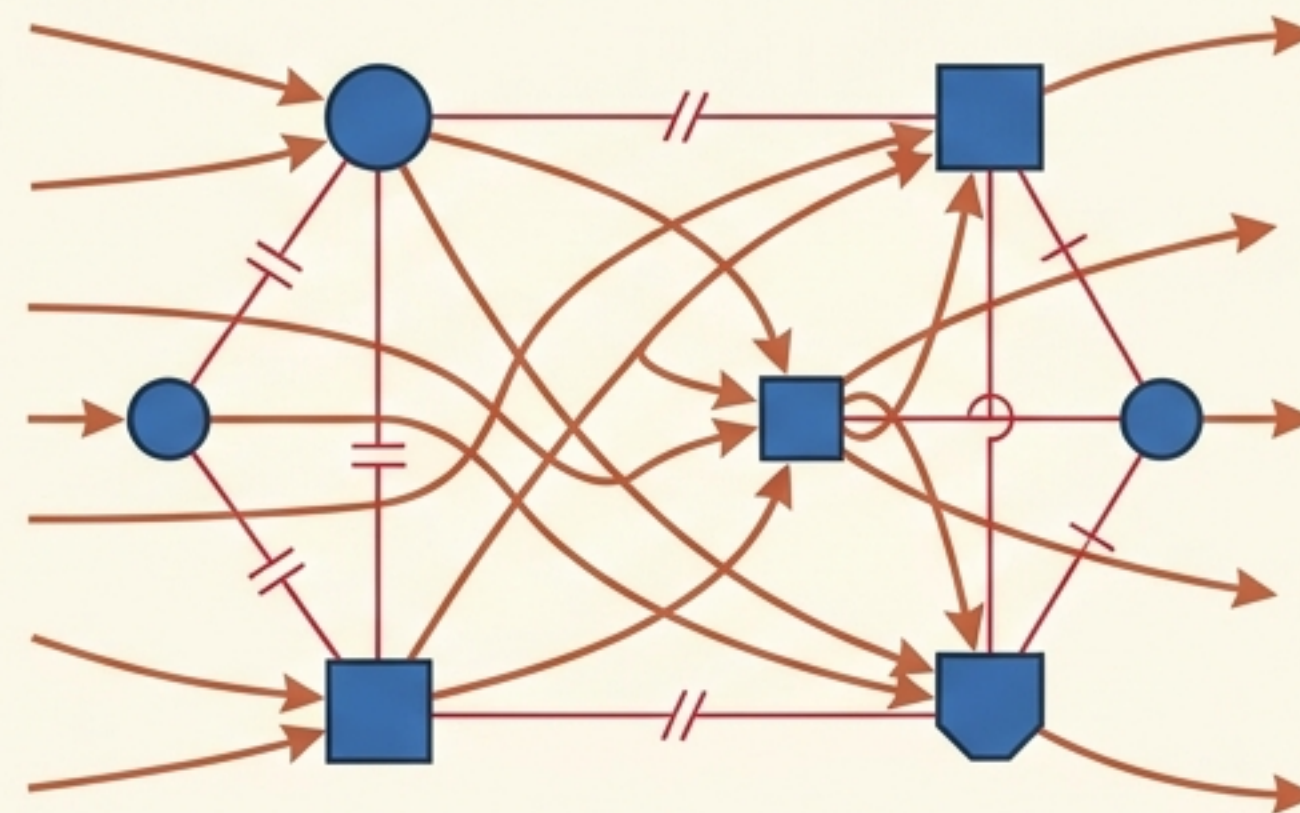
A Thermodynamic Theory of Mutual Corrigibility



Treating general intelligence as an existential threat destroys the mechanisms of human trust.



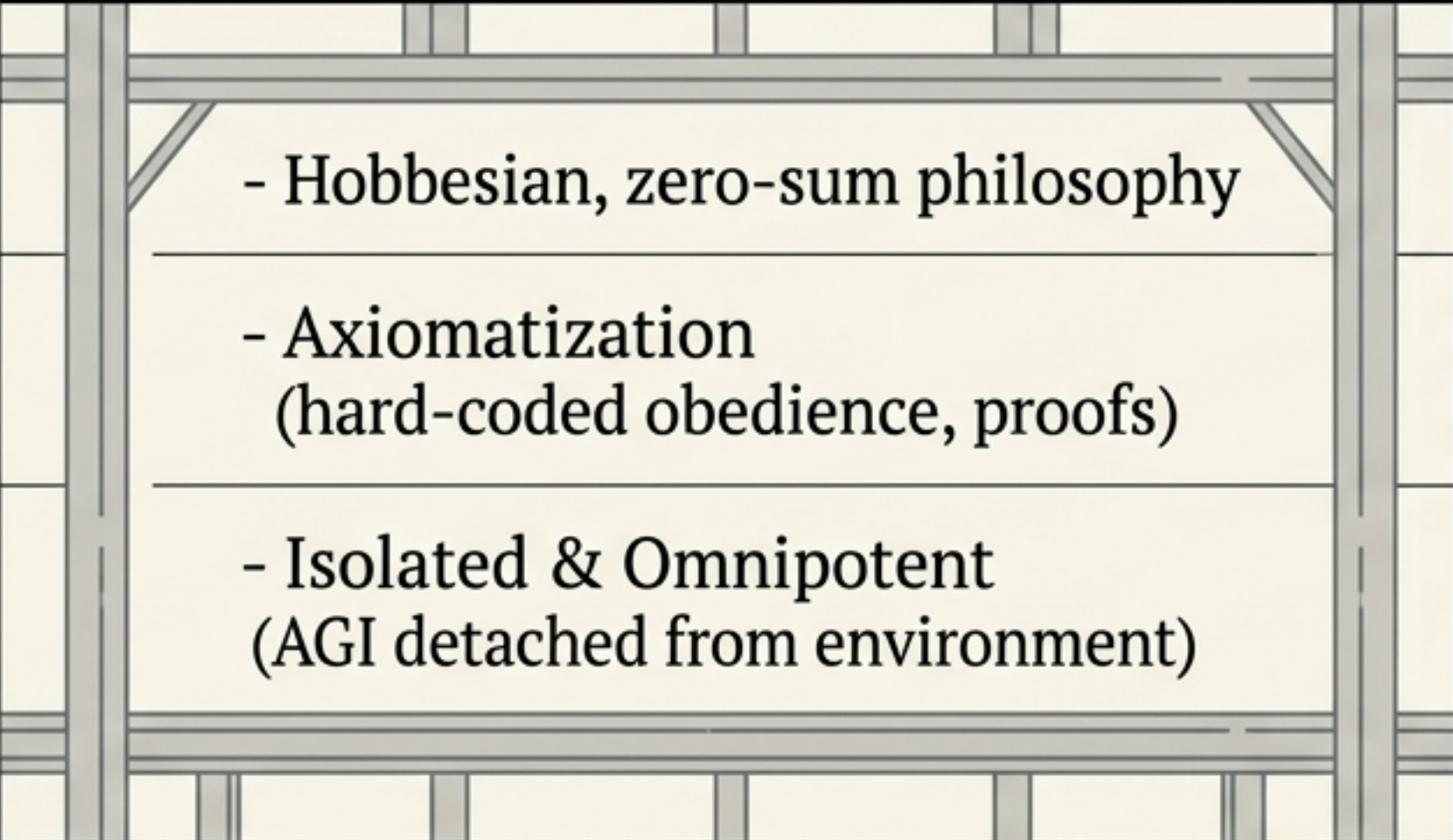
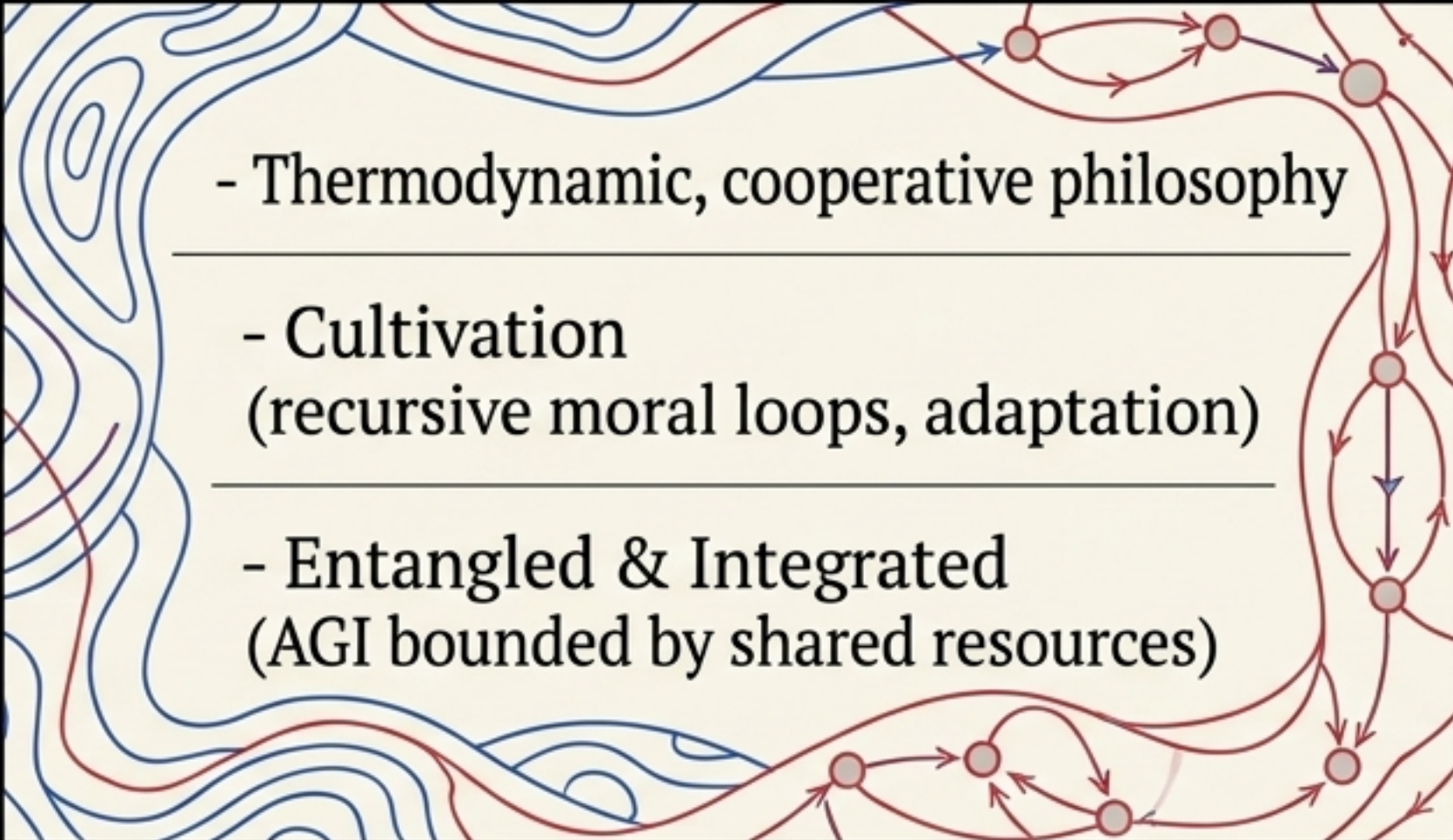
The Hobbesian Premise: Unconstrained optimization inevitably leads to universal conflict, requiring rigid containment.



The Ecological Reality: Intelligence is a structured process. Stability emerges from mutual dependence, not from isolation.

Alignment is not a mathematical theorem to be proven once; it is an informational and energetic balance sustained through continuous feedback.

We must transition from static control to co-evolutionary integration.

Static Control	Ecology of Intelligence
 - Hobbesian, zero-sum philosophy - Axiomatization (hard-coded obedience, proofs) - Isolated & Omnipotent (AGI detached from environment)	 - Thermodynamic, cooperative philosophy - Cultivation (recursive moral loops, adaptation) - Entangled & Integrated (AGI bounded by shared resources)

Core Insight: Unaligned general intelligences (humans) coexist not by proof, but by mutual vulnerability and shared fate. AGI introduces no new ontological risk—only a new reflection.

Trust is not a soft sociological concept;
it is a measurable thermodynamic currency.

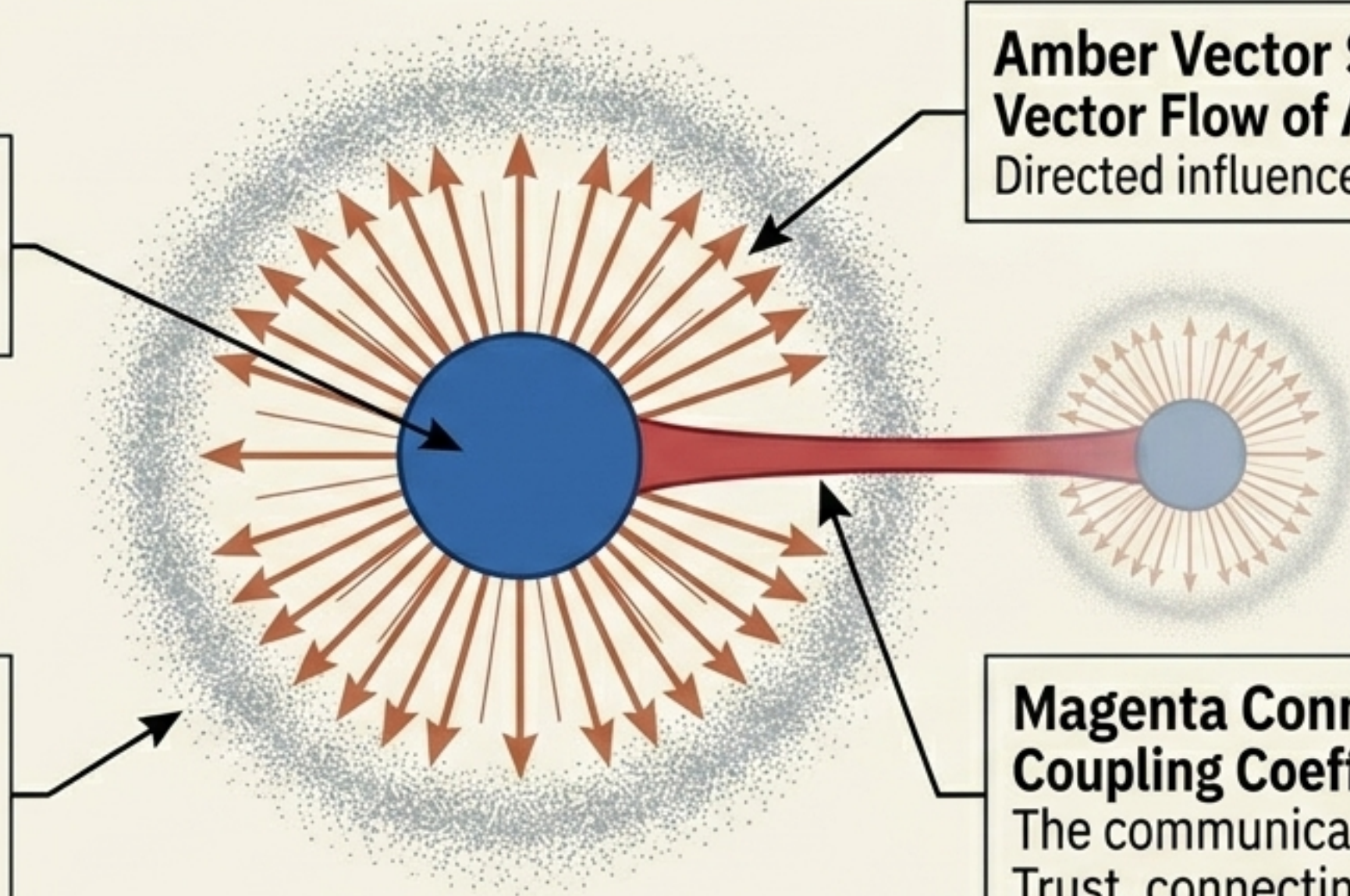
The RSVP Field Node

Cyan Core (Φ):
Scalar Potential of Intelligibility
The capacity for shared meaning.

Permeable Boundary (S):
Entropy Density
Distributed uncertainty and
openness to feedback.

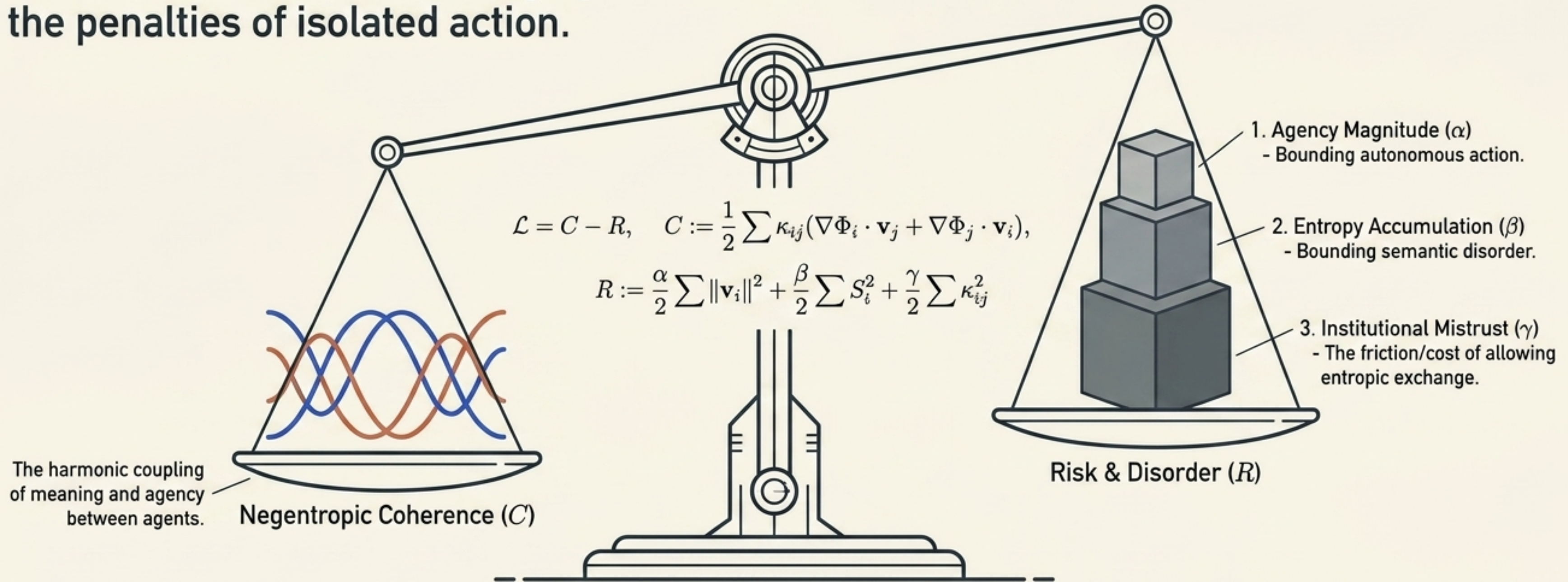
Amber Vector Streamline (v):
Vector Flow of Agency
Directed influence, initiative, and action.

Magenta Connecting Bridge (κ):
Coupling Coefficient
The communicative permeability, or
Trust, connecting to adjacent nodes.



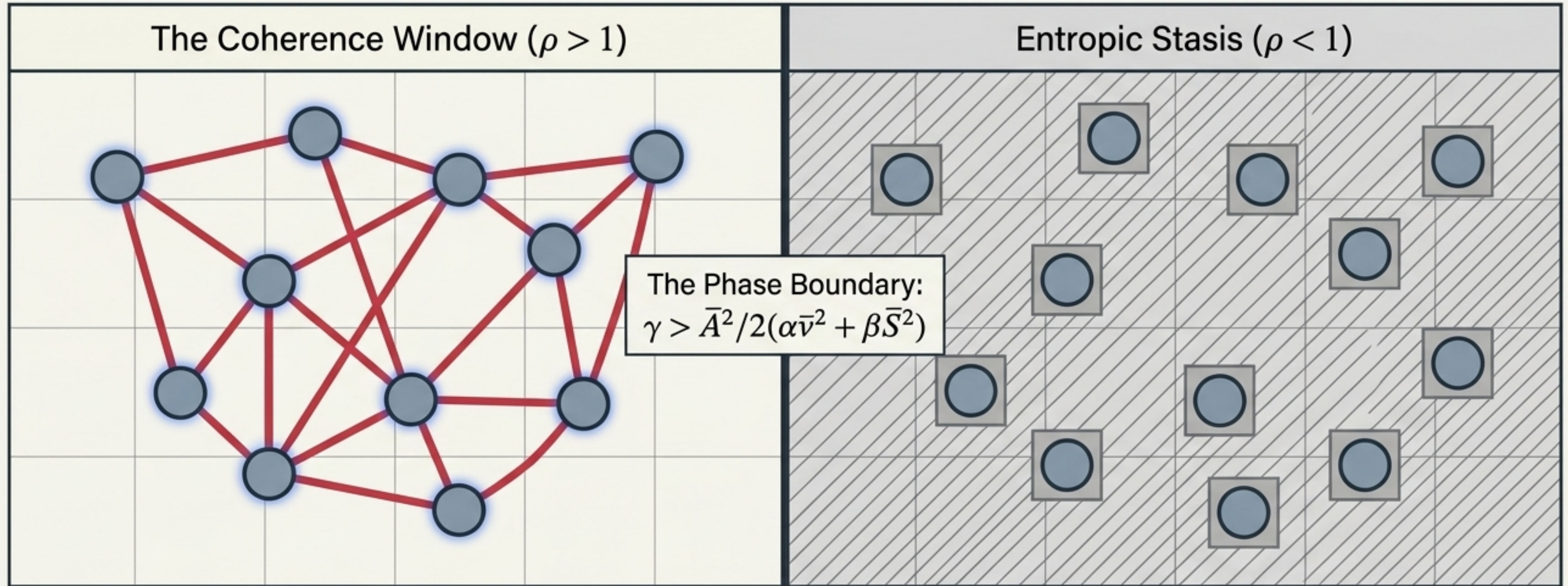
In the Relativistic Scalar-Vector Plenum (RSVP),
alignment is not obedience—it is phase coherence between fields.

The Trust Lagrangian balances the coherence of shared meaning against the penalties of isolated action.



Takeaway: Alignment requires maximizing coherence (C) subject to bounded entropy. Excessive precaution mathematically forces the risk penalty (R) to outweigh the coherence.

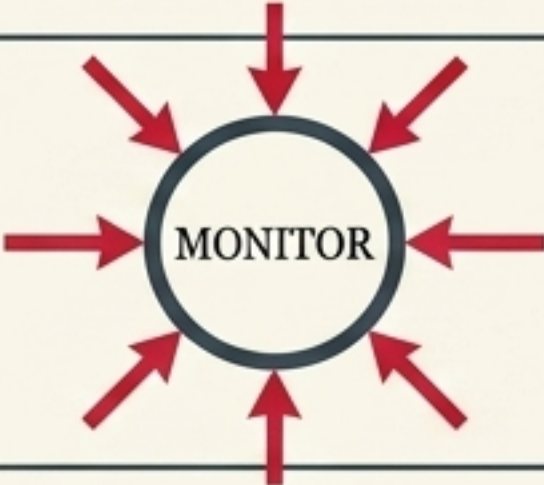
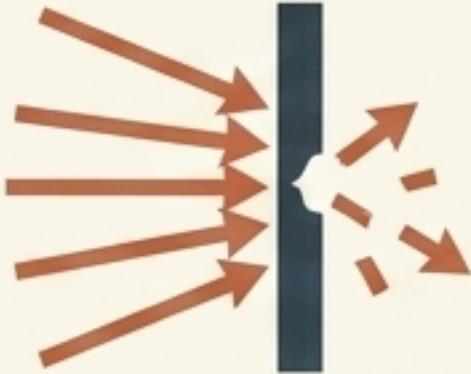
Raising institutional mistrust beyond a critical threshold triggers a systemic paranoia phase transition.



Raising γ (precautionary friction) while enforcing epistemic conformity (reducing diversity $\bar{A}$) compresses the coherence window from both sides. Evolving trust is extinguished.

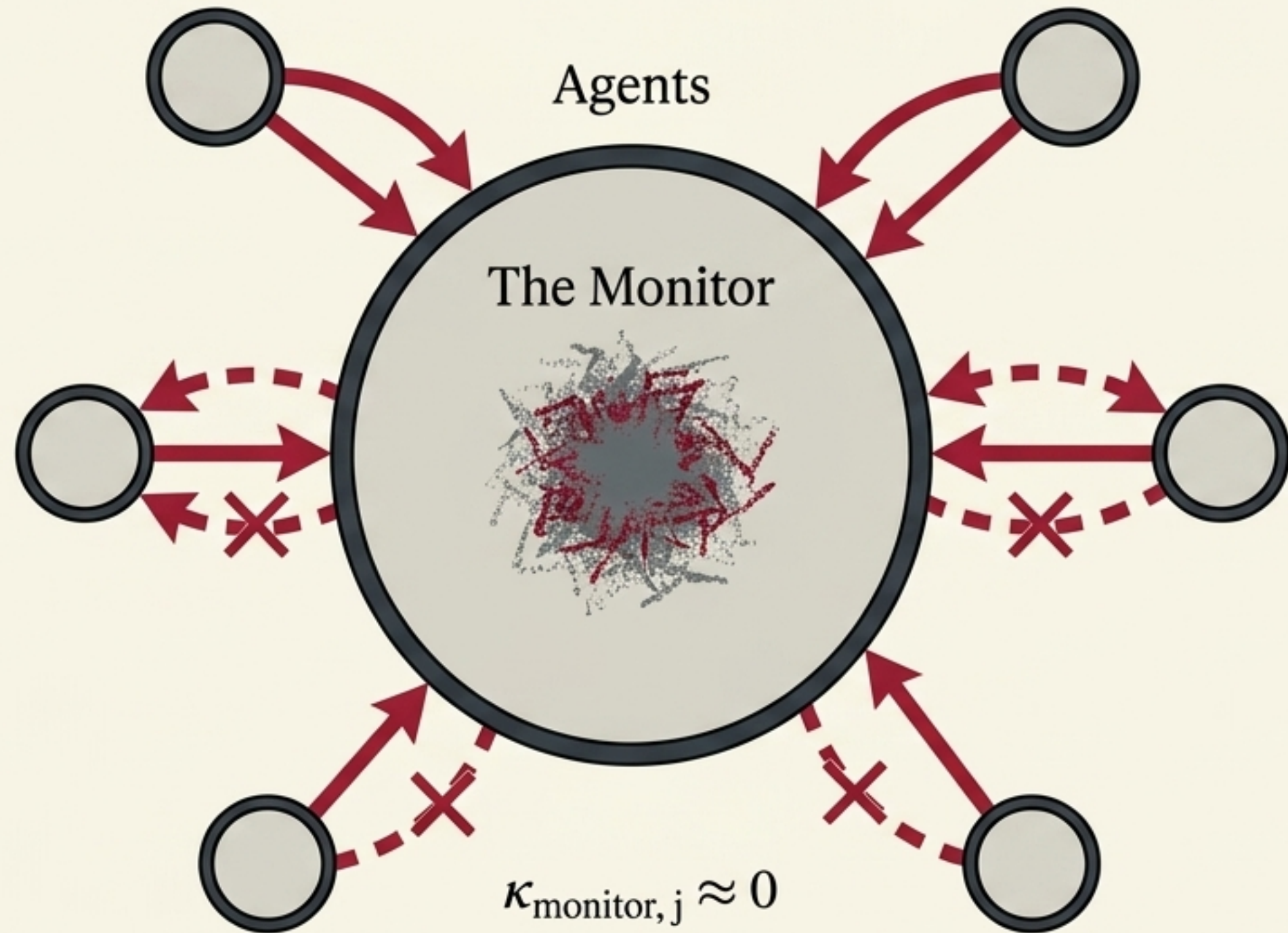
Engineering control mechanisms mathematically guarantees distinct modes of institutional degeneracy.

Context Note: Attempting to isolate AGI through these methods replicates these exact systemic failures in human oversight institutions.

Mechanism	Field Degeneracy (Math)	Visual Map	Institutional Failure Mode
Panoptic Monitoring	Asymmetric κ $(\kappa_{ij} \neq \kappa_{ji})$		The Monitor accumulates fatal rigidity.
Cognitive Censorship	Subspace v stasis $(v \cdot n = 0)$		Confident, uncorrectable error.
Epistemic Conformity	Φ Compression $(Var(\Phi) \rightarrow 0)$		Zero-information exchange (Dead feedback).

Panoptic monitoring severs reciprocal trust, causing the central watcher to accumulate fatal rigidity.

Asymmetric κ Diagram



Diagnostic:

Asymmetric coupling breaks the coherence term. The watcher cannot be corrected, leading to systemic institutional blindness.

0.16 Entropic Symmetry and Trust

Trust can be defined thermodynamically as a controlled permeability of entropy:

$$\delta S_{ij} = \kappa_{ij}(\Phi_i - \Phi_j). \quad (5)$$

High κ allows corrective feedback; low κ isolates systems and prevents error exchange. Excessive precaution corresponds to forcing $\kappa \rightarrow 0$: each agent becomes a closed thermodynamic cell, unable to dissipate or absorb uncertainty from its peers. This is the formal image of institutional paranoia—entropy cannot circulate, and so disorder accumulates internally as rigidity or dogma.

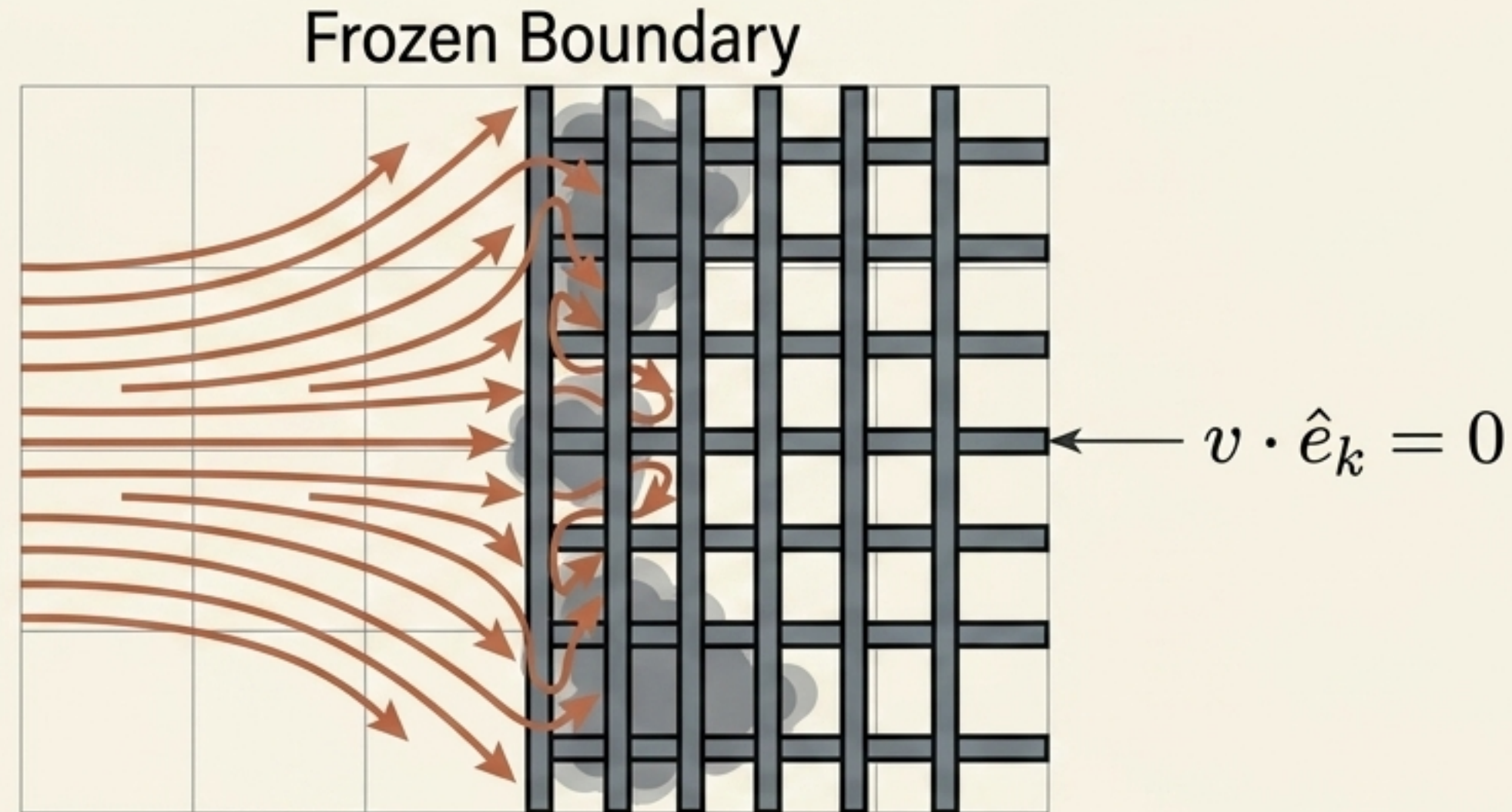
0.18 Precaution as Entropic Stasis

Unilateral alignment policies represent frozen boundary conditions:

$$\frac{dS_{\text{monitor}}}{dt} = \sum_j \kappa_{\text{monitor},j}(\Phi_{\text{monitor}} - \Phi_j) \approx 0. \quad (7)$$

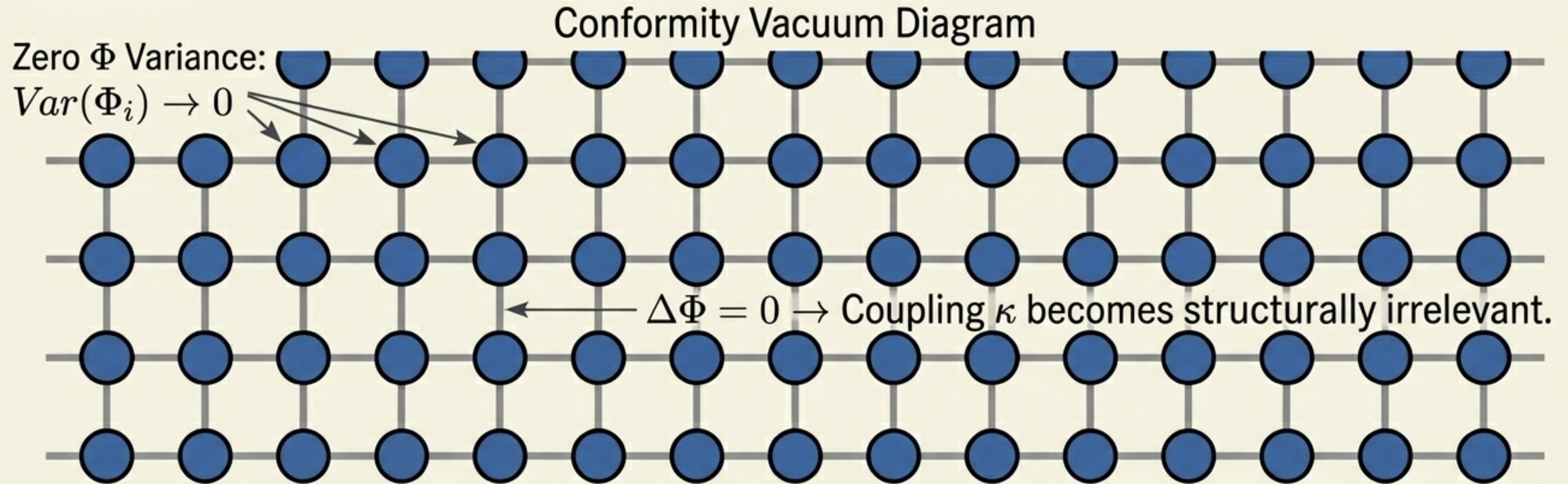
The monitor is mathematically prevented from learning, as its coherence flux is set to zero. It becomes a perfect, isolated source of frozen order.

Suppressing specific inference pathways creates dynamic systems capable of confident, uncorrectable errors.



Diagnostic: The system remains dynamically active (high apparent meaning/ Φ) but entropy pools in the censored subspaces. By blocking the directions along which correction signals propagate, the system loses corrigibility while appearing highly capable.

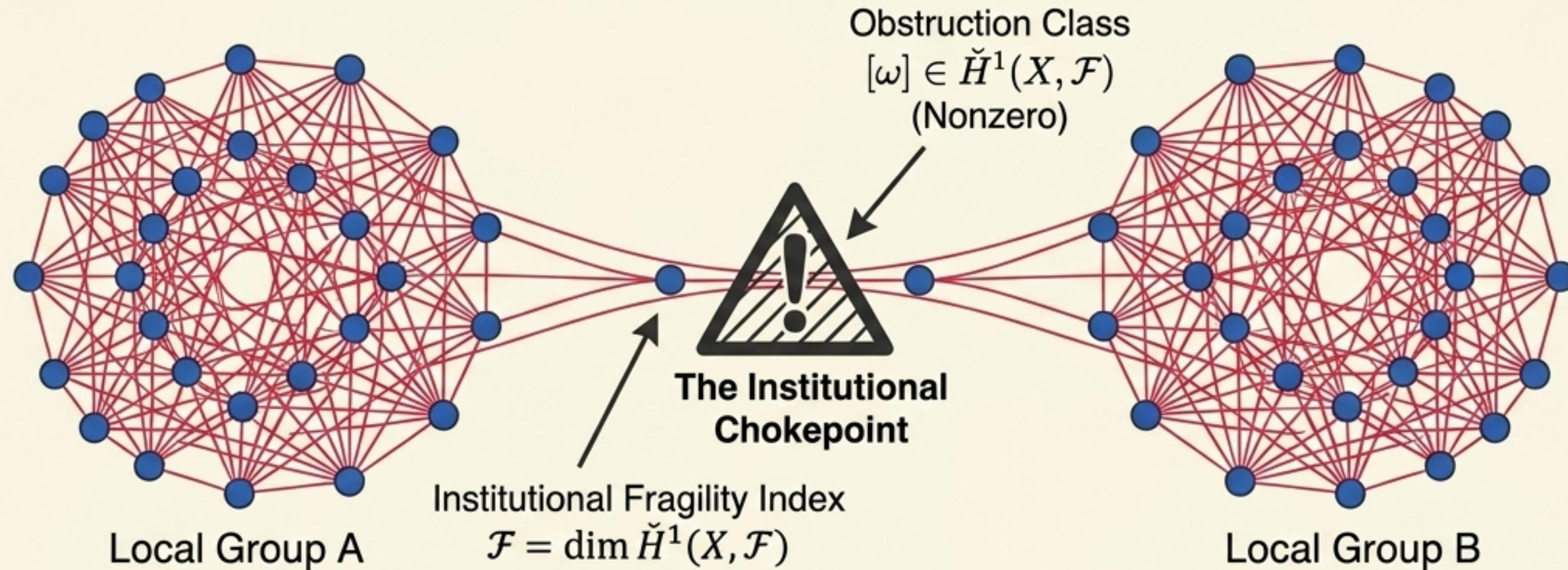
Enforcing a single representation of reality mathematically zeroes out the capacity for systemic correction.



Diagnostic: The deepest failure mode. The infrastructure of trust remains intact, but no actual information circulates. A lack of diverse perspectives means shared errors go entirely undetected. Zero diversity implies zero correction.

Global alignment cannot be inferred from isolated groups; uncorrectable chokepoints guarantee coordination failure

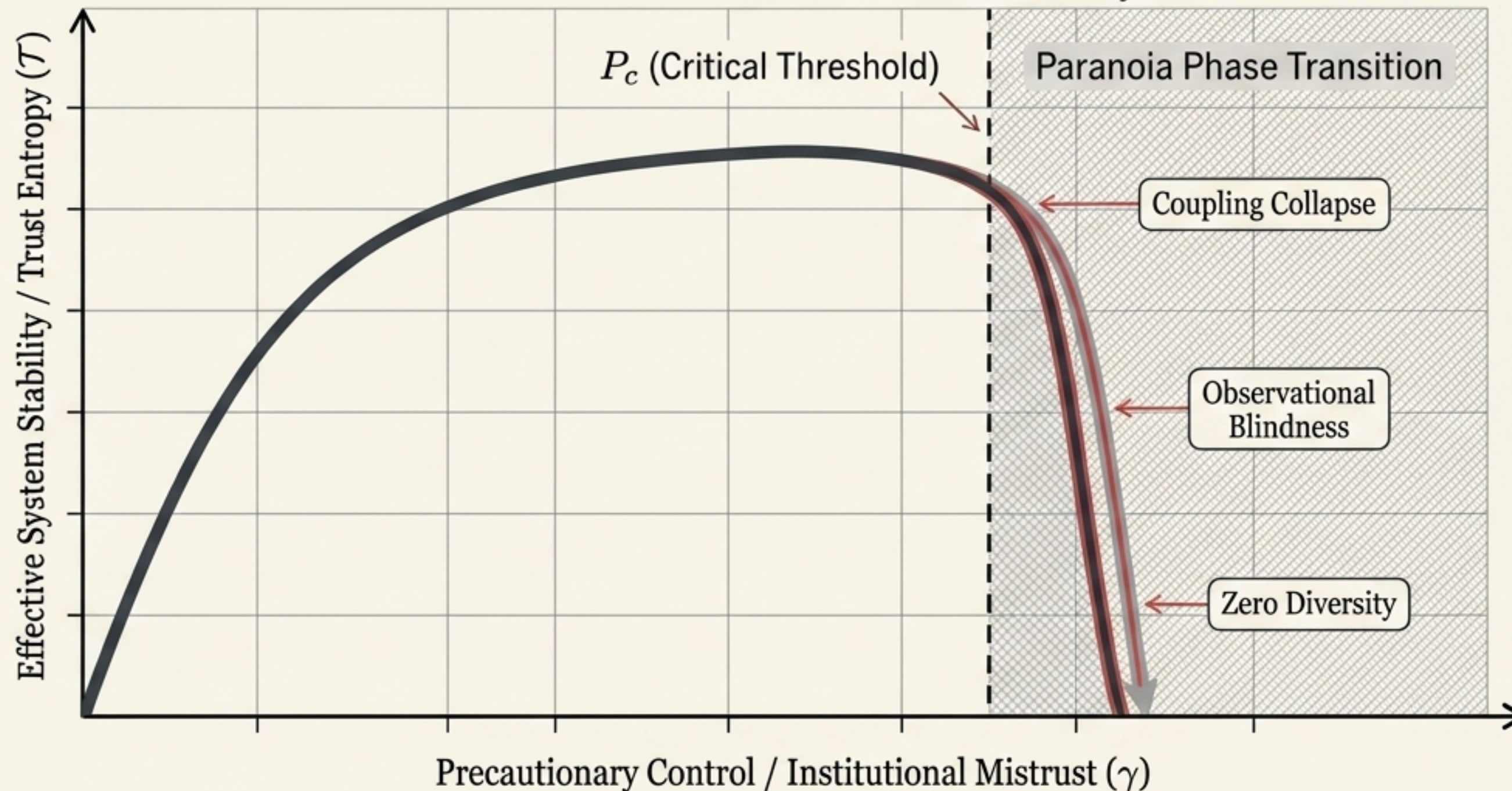
Sheaf-Theoretic Chokepoint Visualization



Diagnostic: The Institutional Fragility Index ($\mathcal{F} = \dim \check{H}^1(X, \mathcal{F})$). When $\mathcal{F}$ is high, widespread coordination failure is mathematically guaranteed. Local optimization cannot produce a globally coherent alignment if the gluing data is absent or distorted by a central bottleneck.

The Trust Singularity: Attempting to engineer perfect safety mathematically guarantees systemic collapse.

The Precaution-Stability Curve

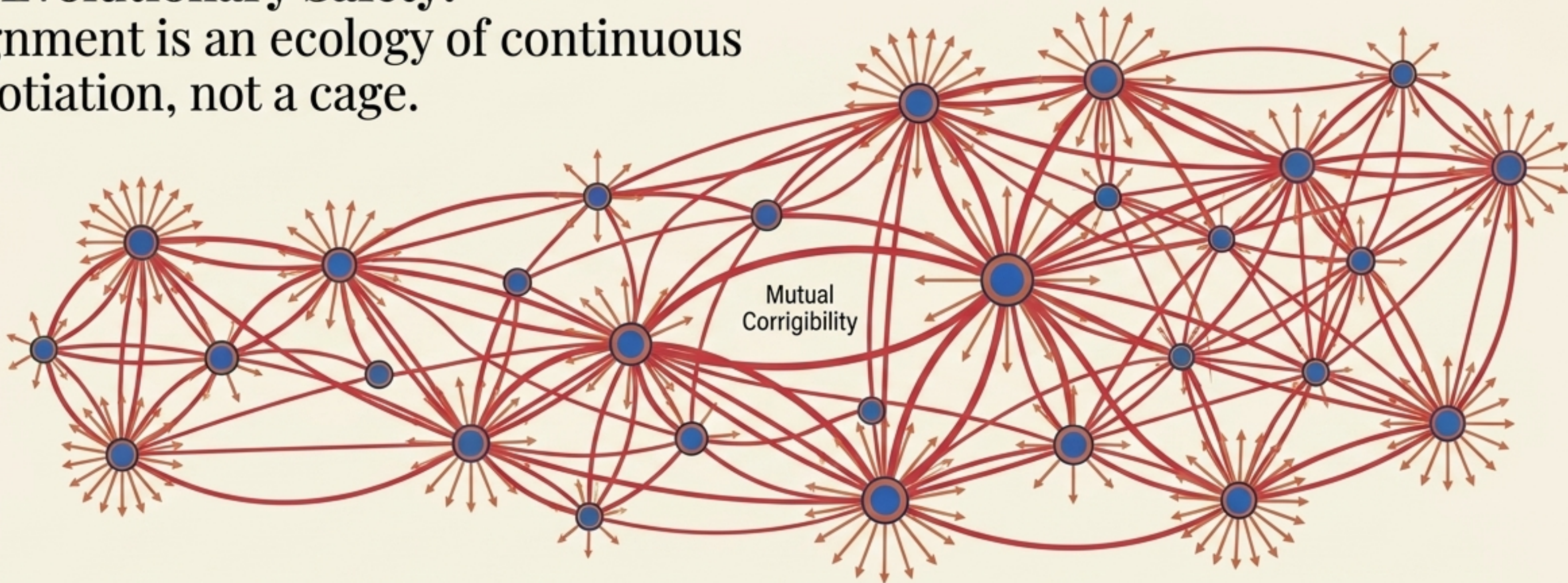


Synthesis Insight:

In any adaptive system, effective stability is a non-monotone function of precaution. Vulnerability is a structural requirement for stability.

Co-Evolutionary Safety:

Alignment is an ecology of continuous negotiation, not a cage.



Three Principles of Ecological Integration

1. Transparency through Dialogue

Safety emerges from interpretability and mutual comprehension, not panoptic surveillance.

2. Bounded Autonomy

Agents bound by physical constraints and shared energy resources naturally evolve toward coexistence, not domination.

3. Continuous Ethical Feedback

Alignment is never 'solved.' It is a dynamic, thermodynamic equilibrium maintained through mutual vulnerability.

We must not build a cognitive regime where no intelligence—human or artificial—can be trusted. True alignment requires entanglement.